

Task

Build an Agentic Inventory Replenishment / Manufacturing Workflow

John's suggestion is to take the **inventory use case** and automate the process using the Agent Bricks agent framework.

The core scenario is:

When inventory goes below a particular threshold, an agent detects it and starts an orchestration workflow.

1. Inventory Monitoring Agent

Create an agent that **monitors warehouse/inventory levels**.

When inventory reaches a **low threshold**, the agent should:

- Detect that inventory is low.
- Analyze the situation.
- Make a prediction that the inventory is going to run out soon.
- Determine/recommend that an order needs to be made.
- Notify the responsible human/supervisor.

John specifically described this as an agent that watches the inventory and, when it is low, makes an analysis and prediction.

2. Human Notification / Approval

The workflow must involve a **human in the loop**.

John specifically mentioned **Sarika as the Production Manager** as an example of the human supervisor.

The agent should send her a notification/message saying, in effect:

The warehouse inventory is low.
This is my observation and recommendation.
Do you want to proceed with the order?

The notification should also provide the **facts behind the recommendation**, so that the human can make the decision.

John suggested that this could potentially be integrated with **Microsoft Teams**.

3. Supervisor / Agent Orchestration

The workflow should use Agent Bricks' agent framework and involve **agent orchestration**.

John specifically mentioned using:

- A **Supervisor Agent**
- A **Genie Agent**
- Other agents as required for the workflow

The exact division of responsibilities between these agents was **not explicitly specified** by John, so we should not assume one yet.

4. Manufacturing Request Agent

When the inventory condition requires action, the workflow should trigger another agent that:

places a manufacturing request.

So the high-level flow includes:

Inventory Agent

- detects low inventory
 - analyzes / predicts shortage
 - recommends action
 - human approval
 - **manufacturing request**
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5. Parts Availability Check

The manufacturing-request workflow should then trigger another agent that:

checks whether all required parts are available.

If all required parts are available, the workflow can proceed.

If parts are **not available**, the workflow should:

place an order for the required parts.

John explicitly described this sequence:

Manufacturing Request

- **Check Parts Availability**
- If unavailable → **Place Parts Order**

6. Human Approval Before Placing the Order

A key requirement from John:

The system should NOT automatically place the order without human approval.

Every time an order needs to be placed, the process should involve the human.

The flow John described is:

Agent detects low inventory

- **Agent makes observation / prediction / recommendation**
- **Human receives notification**
- **Human reviews the facts**
- **Human approves**
- **Agent triggers the next agent**
- **Order is placed**
- **Human is informed that the order has been placed**

7. Feedback to the Agent

John specifically mentioned that when Sarika approves the recommendation:

the feedback goes into the agent

and then the agent triggers another agent that places the order.

So human approval is not just an authorization step; John also explicitly wants that feedback to go back into the agent workflow.

8. Confirmation After Order Placement

After the order is placed, the workflow should **confirm back to Sarika that the order has been placed**.

So the end of the flow is:

Human approves

- **Order placement agent**
 - **Order placed**
 - **Confirmation to human**
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9. Use Agent Bricks for the Implementation

John explicitly asked you to:

Take this workflow and try to create it with the agent framework within Agent Bricks.

He said you can use:

- Any appropriate agent
- Supervisor Agent
- Genie Agent
- Other Agent Bricks capabilities

The workflow should be implemented **within Agent Bricks** rather than just being a conceptual workflow.

10. MLflow Is Required

John explicitly said:

You have to use all the MLflow and all for evals and monitoring.

So MLflow is a required part of the implementation.

The task therefore includes:

- Evaluation
- Monitoring
- MLflow

The transcript does **not** specify exactly which MLflow judges, scorers, datasets, or metrics you must use, so we shouldn't assume those yet.

11. Deploy as a Databricks Application

John explicitly asked you to:

Deploy this as an application within Databricks.

So this is not just an Agent Bricks playground/demo.

The completed workflow needs to be **deployed as a Databricks application.**

12. Data — You Can Start With Mock Data

You asked whether you needed to wait for the data engineering team or create your own data.

John's answer was:

Start with a mock-up.

The reason he gave is that the data engineering team will take some time to build the actual data.

So:

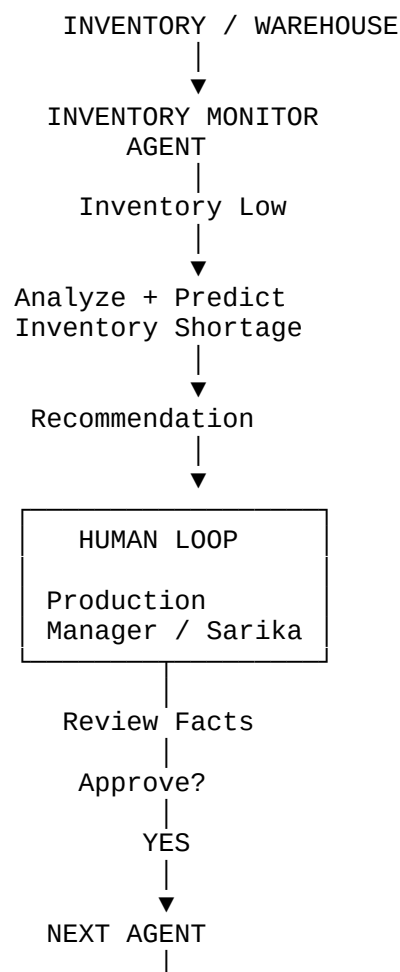
Start with mock data

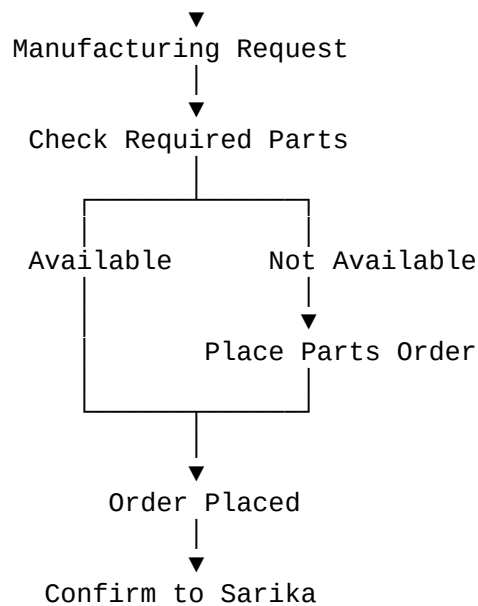
- build the workflow
- when the actual data is ready
- **re-point the workflow to the real data.**

John explicitly approved this approach.

Complete Task Flow

Putting **only John's stated requirements** together:





Required platform elements from John's instructions

Agent Bricks

- Agent(s)
 - Supervisor Agent
 - Genie Agent
- Human-in-the-loop
 - Possibly Microsoft Teams
- MLflow
 - Evaluation
 - Monitoring
- Mock Data initially
 - Re-point to actual data later
- Deploy as Databricks Application

In one sentence

Build an Agent Bricks-based inventory workflow where an agent monitors inventory, predicts an upcoming shortage and recommends replenishment, gets human approval, triggers manufacturing/parts-ordering agents, confirms the completed order, uses MLflow for evaluation and monitoring, starts with mock data, and is deployed as a Databricks application.

What John did *not* specify

To avoid making assumptions, these details are still open:

- Exact number of agents
- Exact responsibility of the Supervisor Agent
- Exact role of the Genie Agent

- Exact inventory threshold
- Exact prediction method
- Exact data schema
- Exact parts/order schema
- Exact Teams integration method
- Exact MLflow evaluation metrics/judges
- Exact Databricks App UI
- Exact approval mechanism
- Exact manufacturing/order APIs

Those should be **design decisions we make next**, rather than treating them as requirements from John.